

Exercise Sheet 10

Discrete Mathematics by Hongfei Fu, 2023.11.9

Name: Chen Zhuoyan Student ID: 523030910037 Class: 2302

1. ([E], Page 133, Exercise 1) Show that the equation

$$f(m, n) = 2^m(2n + 1) - 1$$

defines a one-to-one correspondence between $\mathbb{N} \times \mathbb{N}$ and $\mathbb{N}$.

Answer Area:

- $\forall (m_1, n_1), (m_2, n_2) \in \mathbb{N} \times \mathbb{N}$:
If $(m_1, n_1) = (m_2, n_2)$, then $f(m_1, n_1) = f(m_2, n_2)$.
If $f(m_1, n_1) = f(m_2, n_2)$, then:
Assume that $m_1 \neq n_1$, let's assume that $m_1 > n_1$:

$$2^{m_1}(2n_1 + 1) - 1 = 2^{m_2}(2n_2 + 1) - 1$$

$$2^{m_1-m_2}(2n_1 + 1) = 2n_2 + 1$$

Then the left side of the equation is an even number, but the right side of the equation is an odd number. Contradict. The assumption is false.

So $m_1 = m_2$. Then we have $2n_1 + 1 = 2n_2 + 1$. So $n_1 = n_2$.

$\therefore \forall (m_1, n_1), (m_2, n_2) \in \mathbb{N} \times \mathbb{N}, (m_1, n_1) = (m_2, n_2) \leftrightarrow f(m_1, n_1) = f(m_2, n_2)$. f is an injection.

- $\forall x \in \mathbb{N}$:
If x is an even number, then $\exists m = 0 \in \mathbb{N}, n = \frac{x}{2} \in \mathbb{N}, x = 2^m(2n + 1) - 1$
If x is an odd number, then $\exists m \in \mathbb{N}, x + 1 = 2^m \times k$. (k is an odd number)
So $\exists m \in \mathbb{N}, n = \frac{k-1}{2} \in \mathbb{N}, x = f(m, n)$.
So f is a surjection.

$\therefore$ the equation

$$f(m, n) = 2^m(2n + 1) - 1$$

defines a one-to-one correspondence between $\mathbb{N} \times \mathbb{N}$ and $\mathbb{N}$.

2. ([R], Page 154, Exercise 44(a)(b)) Let f be a function from A to B . Let S and T be subsets of B . Show that

- a) $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$
b) $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$

Answer Area:

a)

- For an arbitrary $x \in f^{-1}(S \cup T), \exists y \in S \cup T, y = f(x)$.
If $y \in S$, then $x \in f^{-1}(S)$.
If $y \in T$, then $x \in f^{-1}(T)$.
 $\therefore x \in f^{-1}(S) \cup f^{-1}(T)$.

- For an arbitrary $x \in f^{-1}(S) \cup f^{-1}(T)$
 If $x \in f^{-1}(S)$, then $\exists y \in S, y = f(x)$.
 If $x \in f^{-1}(T)$, then $\exists y \in T, y = f(x)$.
 $\therefore \exists y \in S \cup T, y = f(x)$.
 $\therefore x \in f^{-1}(S \cup T)$.

b)

- For an arbitrary $x \in f^{-1}(S \cap T), \exists y \in S \cap T, y = f(x)$.
 $\therefore \exists y \in S, y = f(x)$, and $\exists y \in T, y = f(x)$.
 $\therefore x \in f^{-1}(S)$ and $x \in f^{-1}(T)$. So $x \in f^{-1}(S) \cap f^{-1}(T)$.
- For an arbitrary $x \in f^{-1}(S) \cap f^{-1}(T) : x \in f^{-1}(S)$ and $x \in f^{-1}(T)$.
 $\therefore \exists y_1 \in S, y_1 = f(x), \exists y_2 \in T, y_2 = f(x)$.
 $\therefore f$ is a function.
 $\therefore y_1 = y_2, y_1 \in T$.
 $\therefore \exists y_1 \in S \cap T, y_1 = f(x)$.
 $\therefore x \in f^{-1}(S \cap T)$.

$$\therefore f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$$